Gullak

The Money Diaries — a story game that teaches money decisions to Grade 11–12 students

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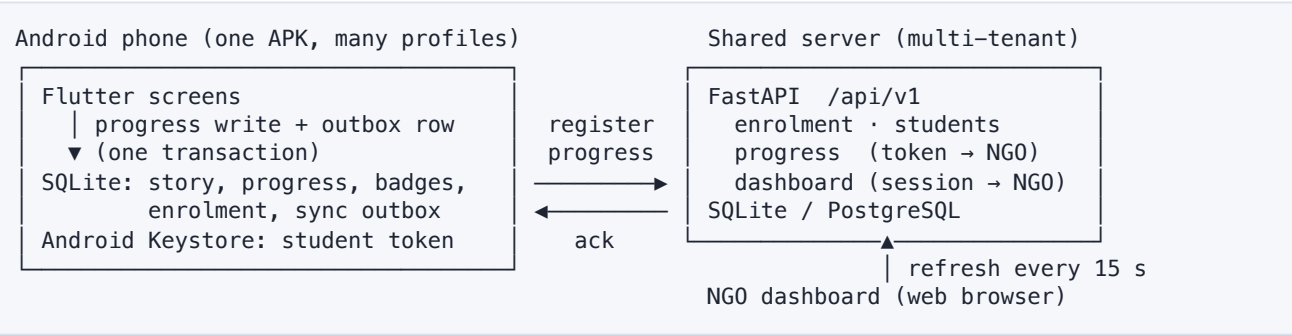
Summary. Gullak is an Android game, in English and Telugu, in which students play a seven-chapter story with pretend money and see their choices carry forward. The game works fully offline. Students may optionally link to their NGO with a code; the app then sends a small progress summary whenever the phone is online, and the NGO sees it on a web dashboard. The app's 158 automated tests pass.

1. Solution

Students follow Asha, a Class 11 student in Hyderabad saving her share of a laptop. Across seven chapters they budget, save, use a bank account, spot scams, weigh credit, compare savings products and handle a family emergency. No real money, login or personal financial information is involved.

The defining design choice is continuity: money and decisions carry from one chapter to the next. Money lost to a scam in Chapter 4 reduces what can be spent in Chapter 5; store credit taken in Chapter 5 keeps charging interest in Chapter 6. Students learn from consequences, and no choice ever blocks the story.

Part	Technology	Role
Android app	Flutter, Dart, Riverpod, SQLite (`sqflite`)	The game; fully offline
NGO API	Python, FastAPI; SQLite locally, PostgreSQL when deployed	Receives progress summaries for any number of NGOs
NGO dashboard	Static HTML, CSS and JavaScript, served by the API	Shows each NGO its own students' progress



2. Approach

One simulation, one state

- Screens send intents to Riverpod controllers, which call pure-Dart rules, which call repositories. Only repositories run SQL, so the rules are testable without a device.
- A single immutable `StoryState` holds every balance, decision and checkpoint; the pure function `StoryRules.apply(state, beat, result)` validates and computes every change.
- Every balance change is also a diary entry, and a test checks that balances always equal the sum of their entries. Outcomes are deterministic and use whole rupees.

Story as data

- Each chapter is a list of beats: dialogue, interactive activity, decision, lesson card and recap. A beat can carry a condition on earlier decisions, which is how the story branches.
- All text sits in one JSON file per language, so new wording or a new language needs no code change.

Reliable persistence

- The SQLite schema (version 5) is built from append-only migrations that run inside a transaction; a failed migration leaves the old data intact.
- The story is saved after every beat with a revision counter, so a stale or repeated save is rejected. Finishing a chapter updates story, progress and badges in one transaction.

Offline-first sync

- A progress change and its queued report are written in the same transaction (an outbox with one row per profile), so the two can never disagree.
- The app uploads on launch, on return to the foreground, after a chapter and on "Sync now". Nothing runs while the app is closed.
- The server accepts a newer revision, ignores a duplicate or older one and rejects conflicting content, so retries on poor networks cannot double-count.

Privacy by design

- No login, real name, phone number or real financial data is collected. NGO sharing is opt-in per profile behind a consent checkbox, and unlinking deletes the server copy.
- The report excludes nicknames, balances and the transaction diary; the server rejects unknown fields.
- Student tokens sit in the Android Keystore and only as hashes on the server. Release builds accept HTTPS only, and no secret is shipped in the app.

Lightweight visuals

All characters, scenes and props are drawn in code with `CustomPainter`. There are no image files, so the app stays small and suits low-end phones.

3. Features

Chapter	Topic	What the student does
1 The First ₹1,000	Budgeting, needs and wants	Builds a budget against a live balance; handles a surprise cost
2 The Laptop Dream	Savings goals	Picks a target and decides how much to save each week
3 Welcome to Digital Money	Banking and UPI	Deposits, withdraws and pays in a working bank screen
4 The Scholarship Message	Scams, OTP and PIN safety	Finds warning signs in four messages; chooses recovery steps
5 Buy Now or Wait?	Debit and credit	Compares options, reads a statement, tries repayment plans
6 Make Your Money Work	Interest and liquidity	Compares 3% and 6%, locks a deposit, meets a mid-term cost
7 When Plans Change	Emergency fund, insurance	Reads a policy and rebuilds the plan for a ₹1,000 clinic bill

Feature	Detail
Two languages	English and Telugu for every screen and all story text, switchable mid-story
Read aloud	Text-to-speech through the phone's own voice engine, with a slow pace option
Progress	Stars scored on careful decisions, never on wealth; five badges; resume after restart; practice replays
Profiles	Several local profiles on a shared phone, each with its own story
Daily check-in	Automatic daily ₹5 pretend reward, streaks with a ₹20 bonus every seventh day, inbox and phone reminders

Accessibility	Respects Android's "remove animations" setting; screen-reader labels; colour always paired with icon and words
NGO enrolment	A student types a centre code once. One APK serves every NGO; a new centre needs no app update
NGO dashboard	Overview, searchable student table, per-centre view, and indicators of where students struggle
Tenant isolation	Every dashboard query is filtered by the coordinator's NGO; another NGO's student returns "not found"
Operations	Command-line tools to add NGOs, centres and codes, revoke or delete students; a Render deployment blueprint

4. Results

158	7 / 7	2	32	~24,000
app tests passing	chapters playable	languages	backend API tests	lines of code and tests

The app suite was run on 11 October 2026: 158 tests passed and 2 were skipped (the optional screenshot renderer, and a test that needs a live backend process on the same machine).

Area	What the tests establish
Game rules	Every chapter's arithmetic, refused inputs, and two complete play-throughs (careful and reckless) with no dead ends
Content	Every text key exists in both languages with matching placeholders
Database	Schema and constraints, upgrades keeping data, a failed migration rolling back, profile isolation
Interface	All seven chapters played through the real screens four times, including Telugu at 130% text on a 360×640 screen with no layout overflow
Sync	Offline play then upload, retries, lost replies, concurrent syncs, revoked tokens, two profiles linked to two NGOs

- **Backend:** 32 API tests cover enrolment, progress revisions, isolation between NGOs and operations; the suite is written to run on SQLite and PostgreSQL.
- **On device:** an end-to-end emulator run was carried out during development: enrol with a code, play a chapter in airplane mode, reconnect, and see the student appear on the correct NGO's dashboard with no manual step.
- **Defect found by testing:** a double-tap test showed a stale button could apply a decision twice; the controller now ignores taps for a beat that is no longer current.

Limitations

- No testing with real students yet, so the effect on learning is unmeasured.
- The Telugu story text (about 540 strings) and the newer interface strings await review by a fluent speaker.
- One difficulty level is playable; Medium and Advanced are shown as "coming later". There are no sound effects.
- All NGOs, codes and students in the repository are fictional; real use needs each NGO's consent process, including guardians of minors.